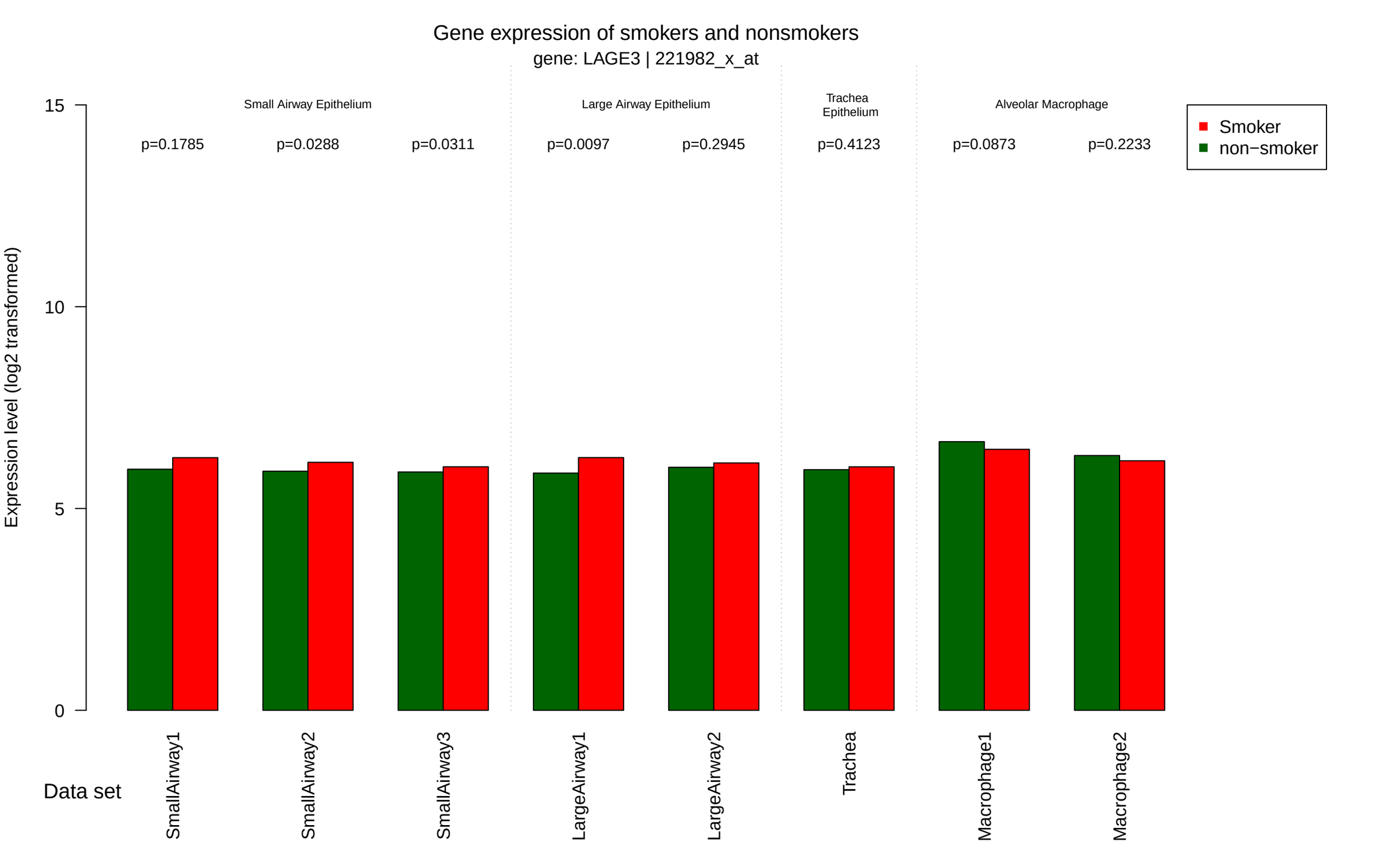
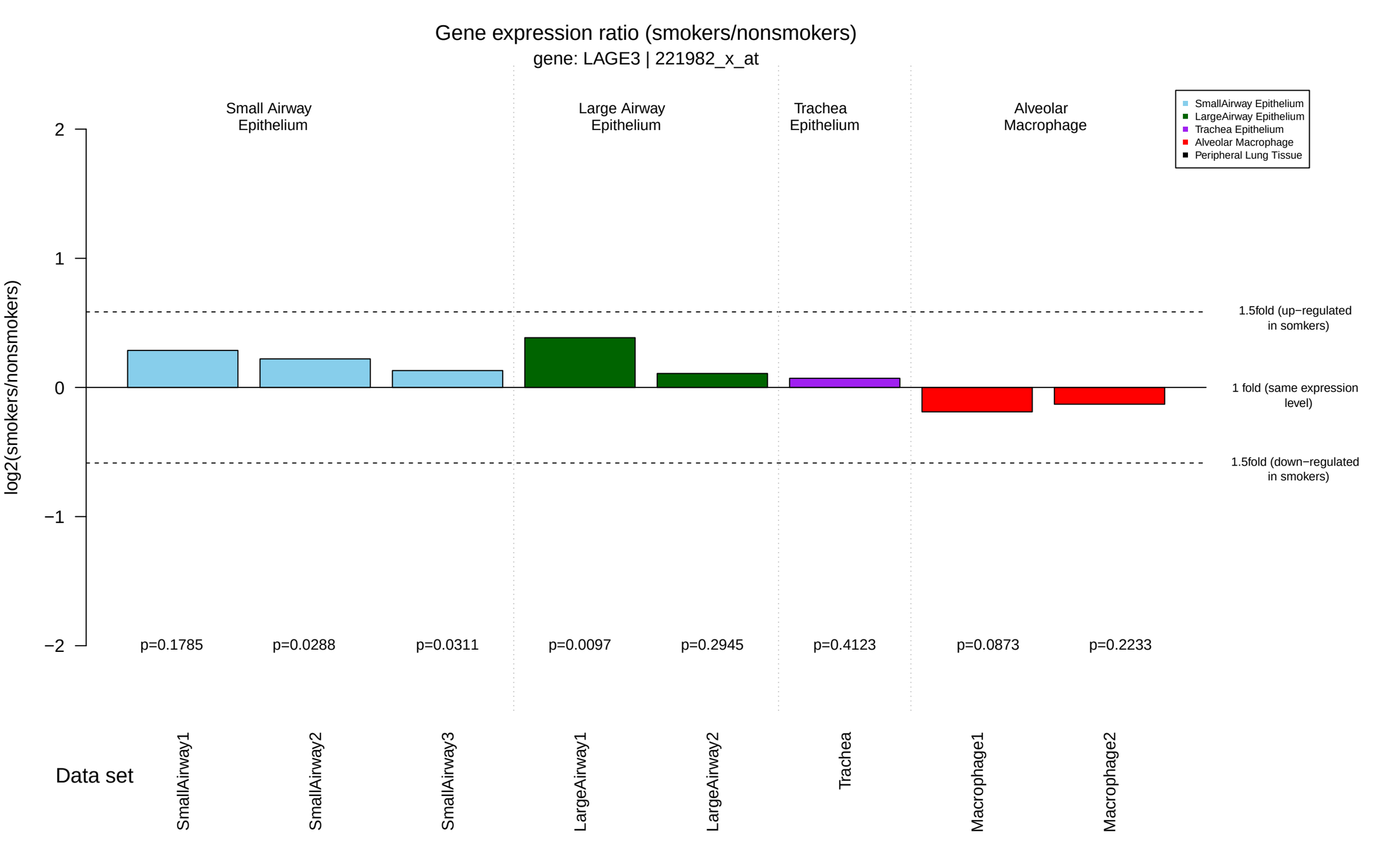
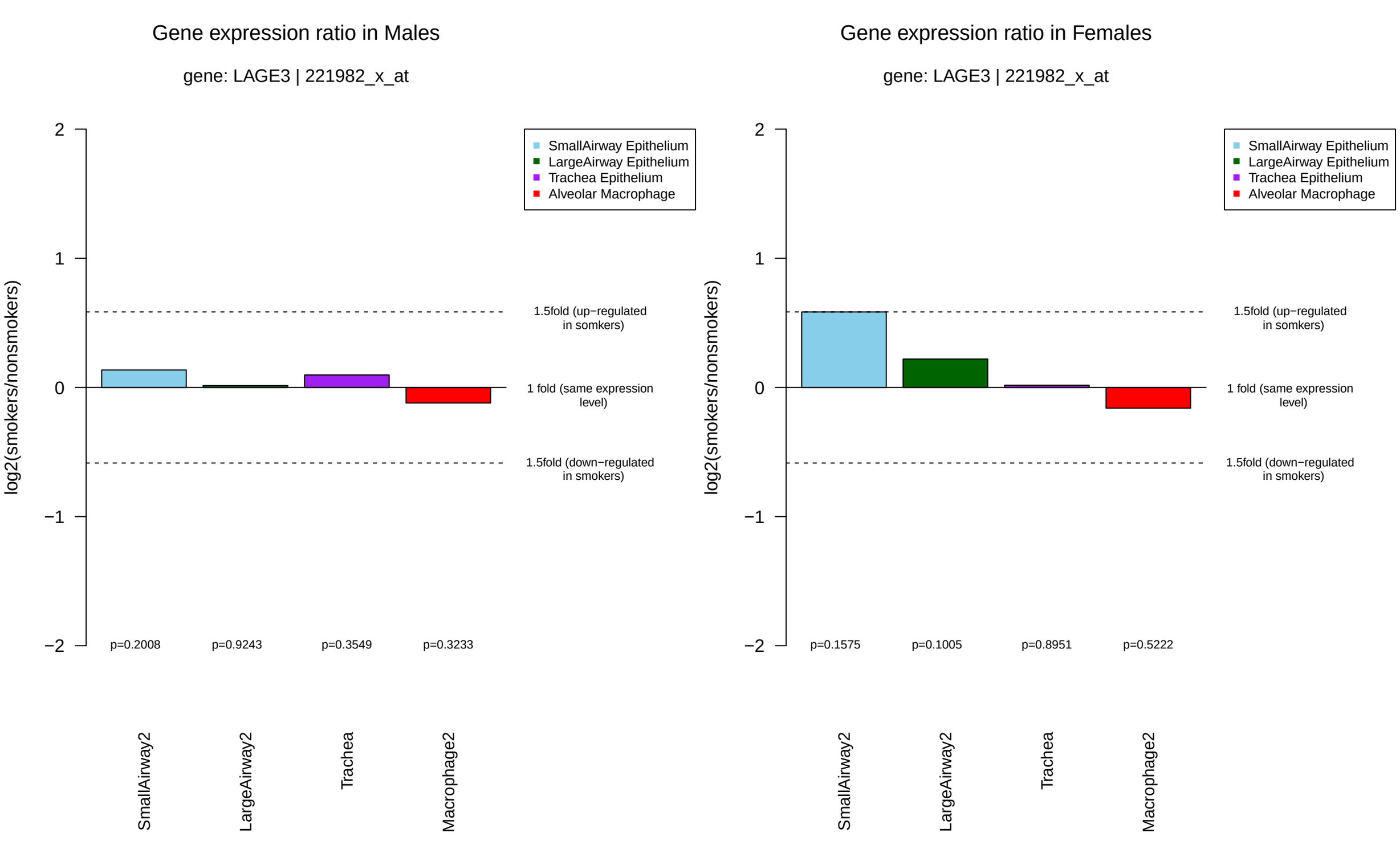
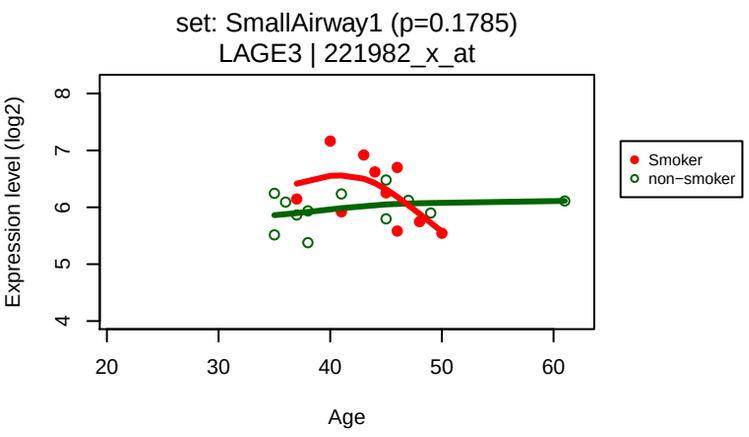
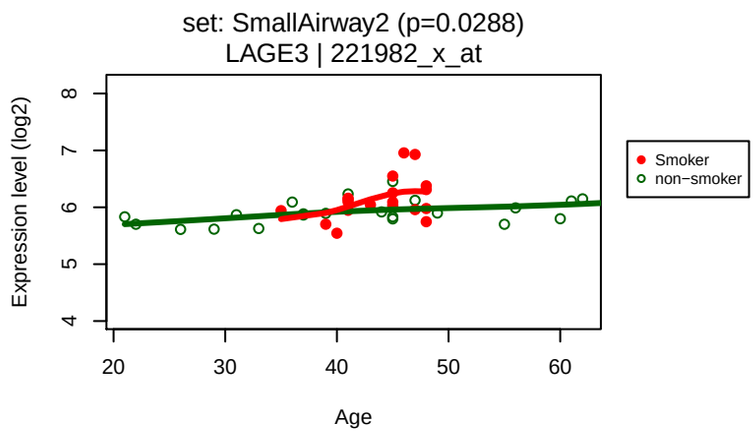
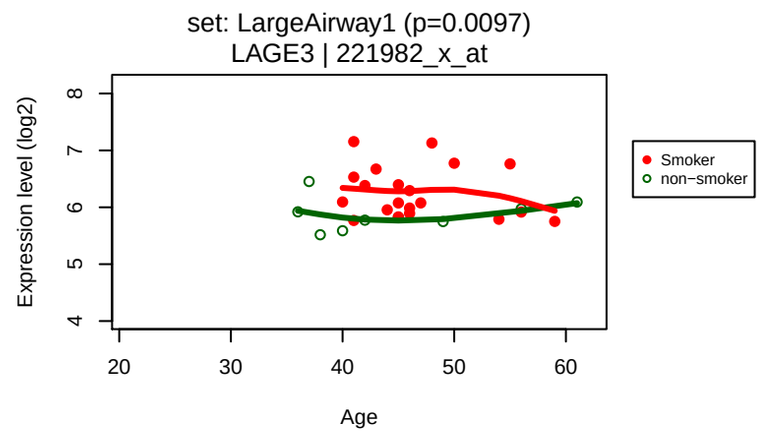

Gene expression of smokers and nonsmokers
gene: LAGE3 | 221982_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1785
p=0.0288
p=0.0311
p=0.0097
p=0.2945
p=0.4123
p=0.0873
p=0.2233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

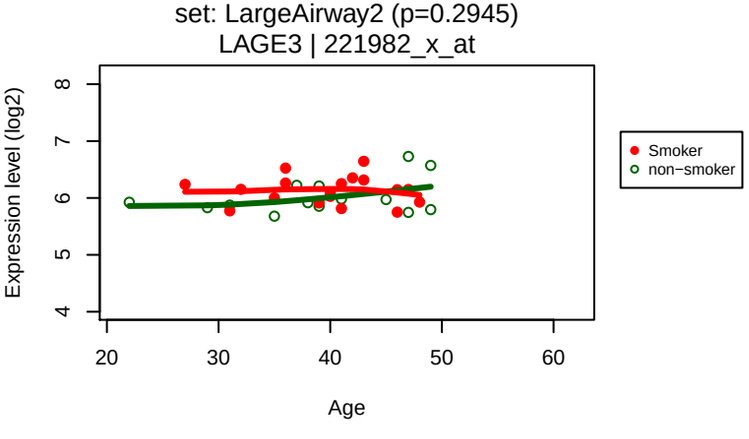

Gene expression ratio (smokers/nonsmokers)
gene: LAGE3 | 221982_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1785
p=0.0288
p=0.0311
p=0.0097
p=0.2945
p=0.4123
p=0.0873
p=0.2233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

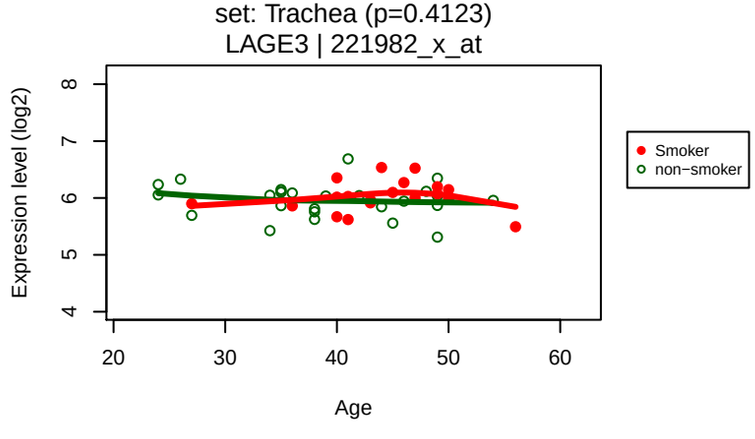

Gene expression ratio in Males
gene: LAGE3 | 221982_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2008
p=0.9243
p=0.3549
p=0.3233
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAGE3 | 221982_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1575
p=0.1005
p=0.8951
p=0.5222
SmallAirway2
LargeAirway2
Trachea
Macrophage2

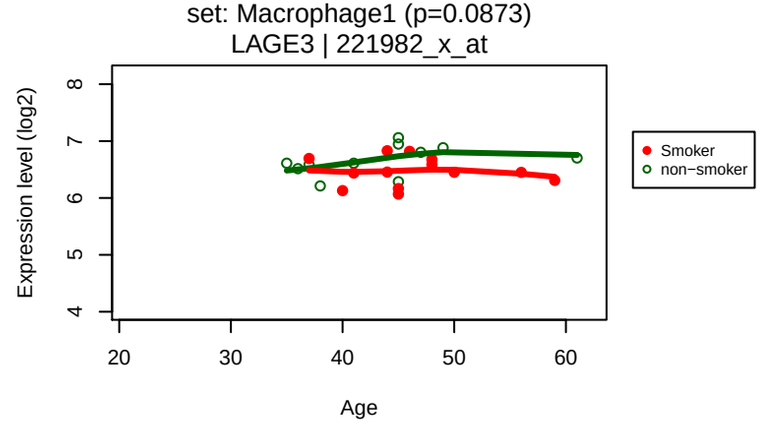

set: SmallAirway1 (p=0.1785)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker

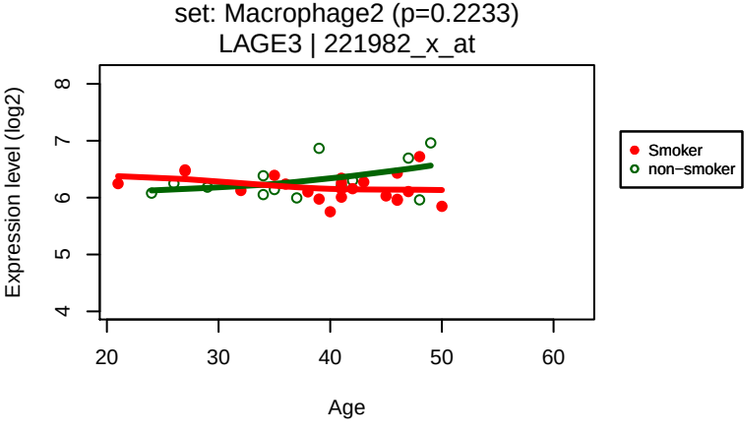

set: SmallAirway2 (p=0.0288)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker

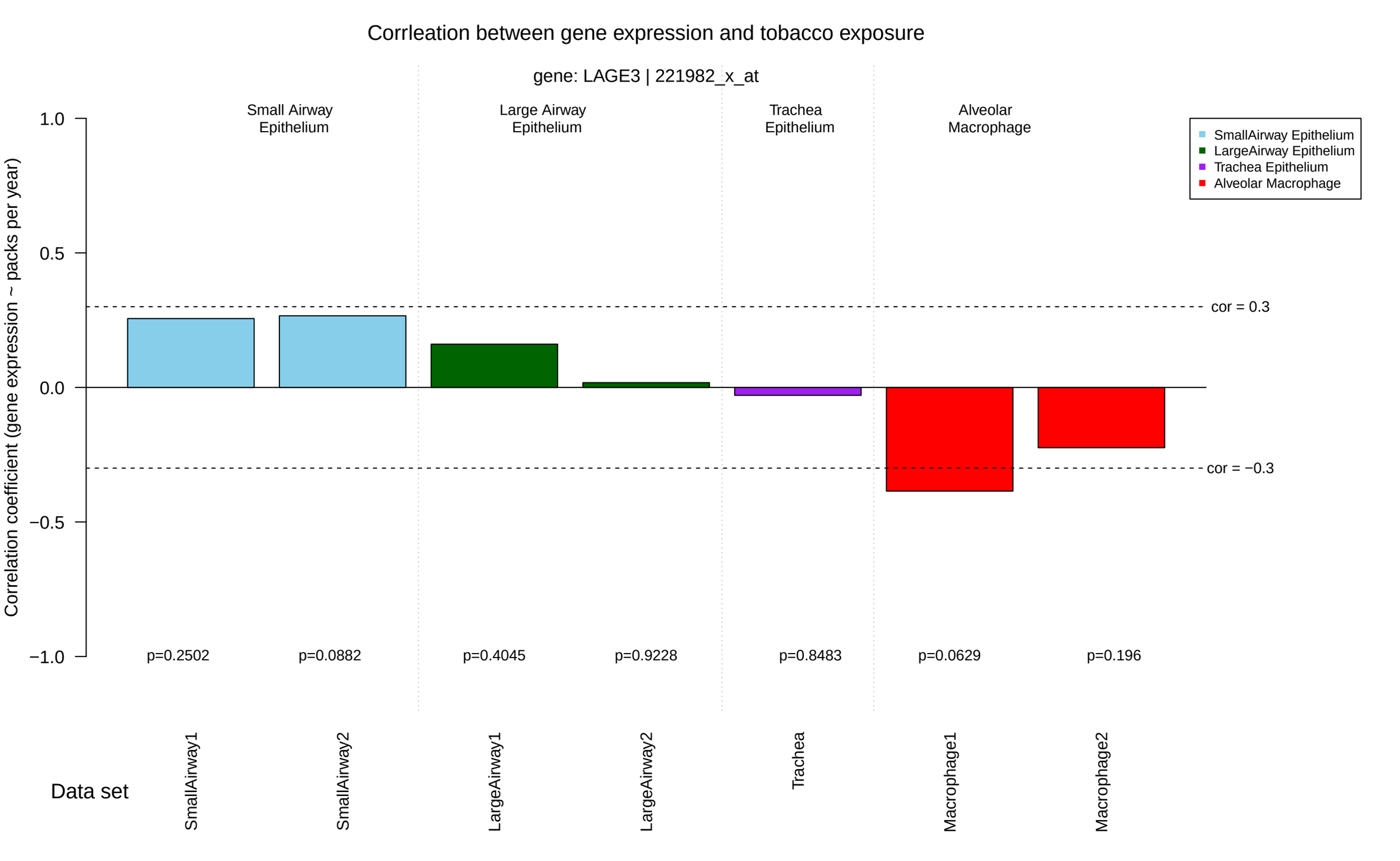

set: LargeAirway1 (p=0.0097)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2945)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4123)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0873)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2233)
LAGE3 | 221982_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAGE3 | 221982_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2502
p=0.0882
p=0.4045
p=0.9228
p=0.8483
p=0.0629
p=0.196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2